

Multivariable Calculus Review

Dealing with two continuous random variables at once requires some multivariable calculus skills.

Each new variable yields a new type of derivative, called a **partial derivative**.

When $f(x, y) : \mathbb{R}^2 \rightarrow \mathbb{R}$, we denote

$$\frac{\partial f}{\partial x}(a, b) = f_x(a, b) = \lim_{x \rightarrow a} \frac{f(x, b) - f(a, b)}{x - a} \quad \text{and} \quad \frac{\partial f}{\partial y}(a, b) = f_y(a, b) = \lim_{y \rightarrow b} \frac{f(a, y) - f(a, b)}{y - b}$$

In practice, we simply treat the variable we are taking the partial derivative with respect to as we normally do while differentiating, and treat the other variable(s) as constant.

For $f(x, y) = x^3y - \cos(xy^2)$ find $f_x, f_y, f_{xx}, f_{xy}, f_{yx}, f_{yy}$

$$\begin{aligned} f_x &= \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} [x^3y - \cos(xy^2)] = 3x^2y + \sin(xy^2) \cdot y^2 = 3x^2y + y^2 \sin(xy^2) \\ f_y &= \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} [x^3y - \cos(xy^2)] = x^3 \cdot 1 + \sin(xy^2) \cdot x \cdot 2y = x^3 + 2xy \sin(xy^2) \\ f_{xx} &= \frac{\partial}{\partial x} [f_x] = \frac{\partial}{\partial x} [3x^2y + y^2 \sin(xy^2)] = 6xy + y^2 \cos(xy^2) \cdot y^2 = 6xy + y^4 \cos(xy^2) \\ f_{xy} &= \frac{\partial}{\partial y} [f_x] = \frac{\partial}{\partial y} [3x^2y + y^2 \sin(xy^2)] = 3x^2 \cdot 1 + 2y \sin(xy^2) + y^2 \cos(xy^2) \cdot x \cdot 2y \\ &= 3x^2 + 2y \sin(xy^2) + 2xy^3 \cos(xy^2) \\ f_{yy} &= \frac{\partial}{\partial y} [f_y] = \frac{\partial}{\partial y} [x^3 + 2xy \sin(xy^2)] = 0 + 2x \sin(xy^2) + 2xy \cos(xy^2) \cdot x \cdot 2y \\ &= 2x \sin(xy^2) + 4x^2y^2 \cos(xy^2) \\ f_{yx} &= \frac{\partial}{\partial x} [f_y] = \frac{\partial}{\partial x} [x^3 + 2xy \sin(xy^2)] = 3x^2 + 2y \sin(xy^2) + 2xy \cos(xy^2) \cdot y^2 \\ &= 3x^2 + 2y \sin(xy^2) + 2xy^3 \cos(xy^2) \end{aligned}$$

Notice that $f_{xy} = f_{yx}$. This is not a coincidence. For a sufficiently smooth function $f(x, y)$, this will always occur, and is known as *Clairaut's Theorem*.

Since there are two ways to take derivatives of $f(x, y)$, there are also two ways to integrate:

We say the partial integral of a such a function is $\int f(x, y) dx$ or $\int f(x, y) dy$

Evaluate both partial integrals of the function $f(x, y) = xe^{xy}$

$$\begin{aligned} \int xe^{xy} dx &= x \cdot \frac{1}{y} e^{xy} - 1 \cdot \frac{1}{y^2} e^{xy} + C \\ &= \left(\frac{xy-1}{y^2} \right) e^{xy} + C(y) \end{aligned}$$

$$\begin{array}{c} + x \\ - 1 \\ 0 \end{array} \begin{array}{c} | \\ | \\ | \end{array} \begin{array}{c} e^{xy} \\ \frac{1}{y} e^{xy} \\ \frac{1}{y^2} e^{xy} \end{array}$$

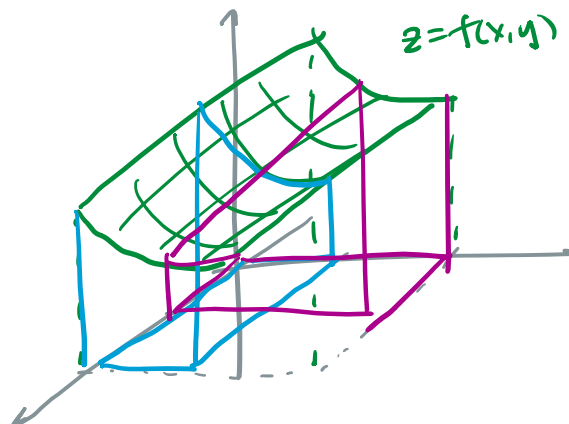
$$\begin{aligned} \int xe^{xy} dy &= x \int e^{xy} dy \\ &= x \cdot \frac{1}{x} e^{xy} + D \\ &= e^{xy} + D(x) \end{aligned}$$

A double integral occurs when you compute two consecutive partial integrals of the same function. The order of the dx and dy terms denote what order to do them in:

$$\begin{aligned} \int_0^1 \left(\int_0^2 (6x^2 + 6y) dx \right) dy &= \int_0^1 2x^3 + 6y \cdot x \Big|_0^2 dy \\ &= \int_0^1 16 + 12y dy \\ &= 16y + 6y^2 \Big|_0^1 \\ &= 16 + 6 = \boxed{22} \end{aligned}$$

$$\begin{aligned} \int_0^1 \left(\int_0^2 (6x^2 + 6y) dy \right) dx &= \int_0^1 6x^2 \cdot y + 3y^2 \Big|_0^2 dx \\ &= \int_0^1 12x^2 + 12 dx \\ &= 4x^3 + 12x \Big|_0^1 \\ &= 4 + 12 = \boxed{16} \end{aligned}$$

These two integrals each represent a volume under $f(x, y) = 6x^2 + 6y$ but over different domains:



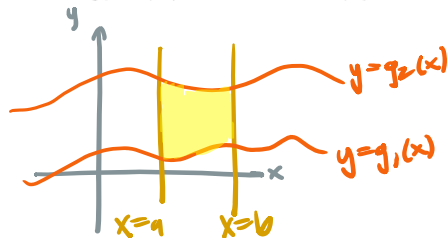
Whenever the domain we are integrating over is a rectangle, $D = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$, we can easily switch between the two orders of integration:

$$\iint_D f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

This is known as *Fubini's Theorem*. It can be useful to simplify certain calculations:

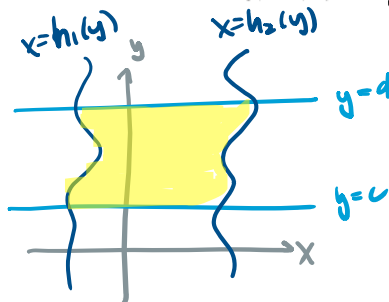
$$\begin{aligned} & \int_0^2 \int_0^1 y^3 e^{xy^2} dy dx \\ &= \int_0^1 \int_0^2 y^3 e^{xy^2} dx dy \\ &= \int_0^1 y^3 \cdot \frac{e^{xy^2}}{y^2} \Big|_0^2 dy \\ &= \int_0^1 y e^{2y^2} - y \cdot 1 dy \\ &= \frac{1}{4} e^{2y^2} - \frac{1}{2} y^2 \Big|_0^1 \\ &= \left(\frac{1}{4} e^2 - \frac{1}{2} \right) - \left(\frac{1}{4} \cdot 1 - 0 \right) = \boxed{\frac{1}{4} e^2 - \frac{3}{4}} \end{aligned}$$

Not all double integrals are conducted over a perfect rectangle. When the domain is of the form $D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ we say it is a **Type 1 region**.



$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

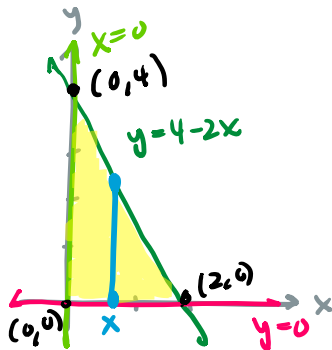
Alternatively, when $D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$ we say it is a **Type 2 region**.



$$\int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

Let D be the triangle between $y = 0$, $x = 0$, and $y = 4 - 2x$.

Evaluate $\iint_D (8x - 2y) dA$ both ways.

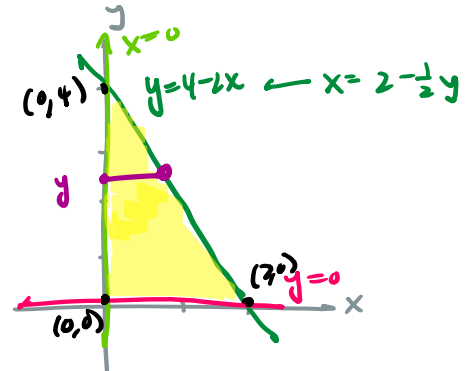


TYPE 1:

$$0 \leq x \leq 2$$

$$0 \leq y \leq 4 - 2x$$

$$\begin{aligned} \iint_D (8x - 2y) dA &= \int_0^2 \int_0^{4-2x} (8x - 2y) dy dx \\ &= \int_0^2 8xy - y^2 \Big|_0^{4-2x} dx \\ &= \int_0^2 8x(4-2x) - (4-2x)^2 dx \\ &= \int_0^2 32x - 16x^2 - (4-2x)^2 dx \\ &= 16x^2 - \frac{16}{3}x^3 - \frac{(4-2x)^3}{3(-2)} \Big|_0^2 \\ &= \left(64 - \frac{128}{3} - 0\right) - \left(0 - 0 + \frac{32}{3}\right) \\ &= \boxed{\frac{32}{3}} \end{aligned}$$



TYPE 2:

$$0 \leq y \leq 4$$

$$0 \leq x \leq 2 - \frac{1}{2}y$$

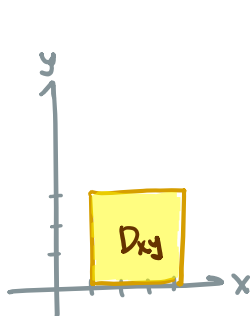
$$\begin{aligned} \iint_D (8x - 2y) dA &= \int_0^4 \int_0^{2-\frac{1}{2}y} (8x - 2y) dx dy \\ &= \int_0^4 4x^2 - 2yx \Big|_0^{2-\frac{1}{2}y} dy \\ &= \int_0^4 4\left(2 - \frac{1}{2}y\right)^2 - 2y\left(2 - \frac{1}{2}y\right) dy \\ &= \int_0^4 4\left(2 - \frac{1}{2}y\right)^2 - 4y + y^2 dy \\ &= 4\left(\frac{2 - \frac{1}{2}y}{3(-\frac{1}{2})}\right)^3 - 2y^2 + \frac{y^3}{3} \Big|_0^4 \\ &= \left(0 - 32 + \frac{64}{3}\right) - \left(-\frac{64}{3} - 0 + 0\right) \\ &= \boxed{\frac{32}{3}} \end{aligned}$$

Notice that on non-rectangular regions, switching the order of integration is not as easy!

While all of your standard integration techniques are still available in double integrals, sometimes it is advantageous to use a more aggressive form of substitution.

For any substitution $x = g(u, v)$ and $y = h(u, v)$, we change our coordinate system from the xy -axis to the uv -axis. This changes our domain, so the limits of integration change.

Try $\int_0^3 \int_1^4 y^2 \sqrt{x} \, dx \, dy$ using $x = u^2$, $y = \sqrt{v}$



$$x = u^2 \rightarrow u = \sqrt{x}$$

$$dx = 2u \, du$$

$$x=1 \rightarrow u=1$$

$$x=4 \rightarrow u=2$$

$$y = \sqrt{v} \leftrightarrow v = y^2$$

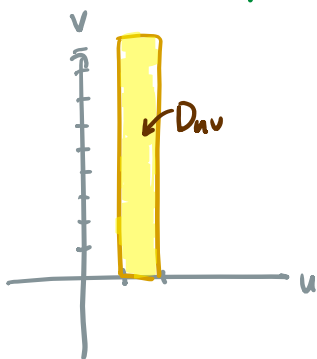
$$dy = \frac{1}{2} v^{-1/2} \, dv$$

$$y=0 \rightarrow v=0$$

$$y=3 \rightarrow v=9$$

$$\int_0^3 \int_1^4 y^2 \sqrt{x} \, dx \, dy = \int_0^9 \int_1^2 v \cdot u \cdot (2u \, du) \left(\frac{1}{2} v^{-1/2} \, dv \right)$$

$$= \int_0^9 \int_1^2 v u \cdot \underbrace{\left(u v^{-1/2} \right)}_{\text{STRETCHING FACTOR}} \cdot du \, dv$$



The scaling factor that stretches dA is called the Jacobian:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$

This means that $\iint_{D_{xy}} f(x, y) \, dA = \iint_{D_{uv}} f(g(u, v), h(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, dA$

Some regions are best to integrate when switched into polar form:

$$x = r \cos(\theta) \text{ and } y = r \sin(\theta)$$

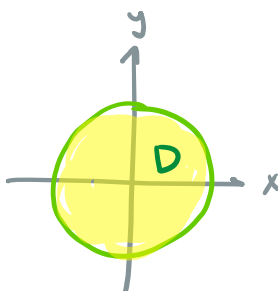
What is the Jacobian in this substitution?

$$J = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = \cos\theta \cdot r\cos\theta - \sin\theta(-r\sin\theta) \\ = r[\cos^2\theta + \sin^2\theta]$$

$$= \underline{\underline{r}}$$

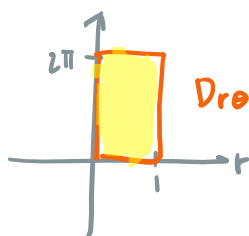
$$dA = dx dy = dy dx = r dr d\theta = r d\theta dr$$

Evaluate $\iint_D e^{x^2+y^2} dA$ where D is the unit circle.



POLAR FORM:

$$0 \leq r \leq 1 \\ 0 \leq \theta \leq 2\pi$$



$$\begin{aligned} &= \int_0^{2\pi} \int_0^1 e^{(r\cos\theta)^2 + (r\sin\theta)^2} \cdot r \cdot dr d\theta \\ &= \int_0^{2\pi} \int_0^1 e^{r^2} \cdot r \cdot dr d\theta \\ &= \int_0^{2\pi} \left. \frac{1}{2} e^{r^2} \right|_0^1 d\theta \\ &= \int_0^{2\pi} \left(\frac{1}{2} e - \frac{1}{2} \right) d\theta \\ &= \left(\frac{1}{2} e - \frac{1}{2} \right) \theta \Big|_0^{2\pi} \\ &= \left(\frac{1}{2} e - \frac{1}{2} \right) \cdot 2\pi = \boxed{\pi(e-1)} \end{aligned}$$